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Making the Invisible Intelligible

*Machine-learning–derived representations of AMOC instability
to improve public understanding of complex climate tipping risks*

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Abstract

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List of Abbreviations

AMOC	Atlantic Meridional Overturning Circulation
EWS	Early Warning Signal
LSTM	Long Short-Term Memory
ML	Machine Learning
PCA	Principal Component Analysis
RAPID	Rapid Climate Change Programme (26°N array)
SDG	Sustainable Development Goals
Sv	Sverdrup ($10^6 \text{ m}^3/\text{s}$)
IF	Isolation Forest

Part I

Foundations: The Context, The Problem And The System

Chapter 1

Introduction

1.1 The Problem of Knowing What We Cannot See

The systems that most powerfully shape the twenty-first century are, with few exceptions, invisible or hard to fully grasp, even by experts. The slow acidification of the ocean, the silent collapse of insect populations, the redistribution of heat through the atmosphere, the subterranean dynamics of the cryosphere: these are processes that unfold across timescales longer than careers and spatial scales larger than continents. They are measured by instruments that the public does not see, recorded as numbers the public does not read, and projected by models the public is asked to trust without ever being shown how those models think. The defining cognitive condition of the present is not, as is sometimes claimed, a shortage of information. It is a chronic mismatch between the structure of the systems that govern our future and the structure of the minds that must reckon with them.

The mismatch is not incidental. It is well documented. Stoknes (2015) identifies five psychological barriers that systematically blunt the public's engagement with climate information — distance, doom, dissonance, denial and identity — and shows that the conventional formats of climate communication (reports, line charts, projection maps) tend to reinforce rather than soften these barriers. Stoknes works in the tradition of cognitive psychology applied to climate risk, a field that itself traces back to Kahneman and Tversky (1979) and their work on heuristic biases in decision-making under risk. The core idea is that the human brain is not wired to process slow, distant, probabilistic risks the same way it processes an immediate, concrete danger. This is precisely why your work does not stop at analyzing RAPID data: if science alone were enough to produce understanding, the gap Stoknes (2015) describes would not exist.¹ Van Der Linden (2015) have shown empirically that climate risk is perceived as psychologically distant on three axes simultaneously — spatial, temporal and social — and that this triple distance reduces both cognitive engagement and willingness to act. Spence et al. (2012) found the same pattern in a representative survey of the British population: increasing the salience of psychological proximity raised concern; the same information presented at a remove did not.²

1 The main outcome of this thesis, named *Living Atlantic*, is a concrete attempt to work around that bias, not by explaining more but by making the scale and urgency of the phenomenon felt through a perceptual channel different from the purely analytical one.

2 The early warning signals this thesis searched for, namely rising variance and rising autocorrelation before a critical transition (Scheffer et al., 2009), are structurally the same phenomenon as psychological distance as characterized by Spence et al. (2012): both are weak signals, hard to notice unless you know what to look for, that precede an important state change. In the ocean system, the weak signal is statistical and buried in depth data. In

The deeper problem is structural, and it predates the climate debate. Sterman (2008), in a body of work spanning two decades, has shown that even mathematically literate adults struggle systematically to reason about systems with stocks and flows, with delays, with feedback. When asked to predict the trajectory of a quantity governed by simple accumulation, educated participants make errors that betray a deep cognitive tendency to confuse rates with levels, inflows with stocks, and to treat dynamic systems as if they were static — as if a graph of CO₂ emissions were a graph of CO₂ concentration, which it is not. The error is not one of arithmetic; it is one of representation. The mind has not been given a form in which the dynamic structure of the system can be held. And what cannot be held cannot be acted on.

This thesis takes as its starting point the proposition that the public understanding of climate risk is not an educational problem to be solved by more information, nor a communication problem to be solved by better wording. It is a *representational* problem: a problem of finding forms in which the dynamic, multi-scale, often non-linear structure of climate systems can be perceived and inhabited, not merely read.

1.2 The Gap: Why Information Does Not Directly Translate to Understanding

The empirical literature on climate visualisation has matured considerably over the past decade, and its central finding is sobering. Harold et al. (2016) reviewed the state of climate-data visualisation against what is known from cognitive and perceptual science about how non-experts process visual information. Their conclusion is that the formats most commonly used to communicate climate data — line graphs of global mean temperature, choropleth maps of projected change, ensemble spaghetti plots — are mismatched to the cognitive resources of the audiences they are meant to inform. They assume a comfort with abstraction, a familiarity with uncertainty representation, and an internal capacity to mentally animate static frames into dynamic processes that the general public does not, in fact, possess.

Three difficulties compound. The first is the difficulty of *dynamic systems*: as Sterman (2008) has shown, static visualisations of inherently dynamic processes systematically mislead, because the reader sees a curve where the system is a flow. The second is the difficulty of *non-linearity*: tipping elements — thresholds beyond which a system reorganises into a different stable regime — violate the linear intuition that small causes produce small effects, and the visualisation conventions inherited from the natural sciences are poorly equipped to convey the discontinuity that defines them (Armstrong McKay et al., 2022; Lenton et al., 2008). The third is the difficulty of *uncertainty*: the error bar, the shaded envelope, the ensemble spread — the standard graphical conventions for communicating uncertainty — are processed by non-expert readers in ways that often invert their intended meaning, treating uncertainty as evidence of disagreement or unreliability rather than as the structural feature of a forecast that it actually is (Spiegelhalter, 2017).

The result is a paradox the climate communication community has long recognised but not the human psyche, the weak signal is the apathetic indifference that precedes action, or its absence.

resolved. The quantity of climate information available to the public has never been greater, and the scientific consensus on the main contours of climate change has never been firmer. Yet the public's grasp of the *behaviour* of the climate system — its non-linearity, its inertia, its capacity for abrupt reorganisation — has not kept pace. The information has been delivered. The understanding has not.

This gap is the territory of this thesis. The argument advanced here is that closing it will require not more information, nor better-worded information, but information presented in *different representational forms*: forms that exploit the perceptual and embodied resources of the human reader, forms in which the structure of the system is encoded into the structure of the experience. The claim is not that experiential representation is a universal solution; it is that it is an under-explored region of the representational space that deserves rigorous, comparative empirical study. Whether it works — and under what conditions — is the question this thesis sets out to address.

1.3 The Case Study: AMOC as a Stress-Test

The argument of §1.1 and §1.2 is general; the work of this thesis is necessarily specific. To investigate whether experiential representations of complex climate systems can improve public understanding, the thesis takes a single system as its case study: the Atlantic Meridional Overturning Circulation, hereafter the AMOC. The choice is not arbitrary. A system chosen to stress-test the representational problem posed in the preceding sections must exhibit, in extreme form, every property that makes such problems hard. The AMOC exhibits all of them at once. It is literally invisible, in the sense that no surface signature is legible to the naked eye and no everyday experience gives access to it. It is spatially distributed at a basin scale, and its behaviour cannot be inferred from any single location. It operates on multidecadal timescales that exceed human experience and, in the observational record, exceed the record itself. It is non-linear in the strong sense: a candidate tipping element whose weakening could organise itself into an alternative circulation regime with hysteresis (Armstrong McKay et al., 2022; Lenton et al., 2008). And it has civilisational stakes: a substantial slowdown would reorganise the heat budget of the North Atlantic and, with it, the climate of Europe (Buckley & Marshall, 2016).

The AMOC is the meridional component of the North Atlantic circulation that transports warm surface waters northward and returns colder, denser waters southward at depth, sustained by density contrasts of thermohaline origin.³ Its strength is conventionally measured in Sverdrup ($1 \text{ Sv} = 10^6 \text{ m}^3 \text{ s}^{-1}$), and since April 2004 has been observed continuously by the RAPID-MOCHA-WBTS array at 26.5° north (Ben I Moat et al., 2026; Frajka-Williams et al., 2019). This observational record — roughly two decades in a system with variability at multidecadal periods — is the empirical substrate of Part I. It is complemented by the Copernicus Marine Service GREP reanalysis ensemble, which extends the analysis back into the

³ The physical apparatus of the circulation — its governing equations, the Stommel two-box model of bistability, and the derivation of meridional transport as a volume integral — is developed in Chapter 4. This chapter treats the AMOC only insofar as it defines the representational problem.

pre-observational period at the cost of introducing ensemble spread as an epistemic quantity in its own right.

The scientific case for treating the AMOC as a representational priority is not, however, only that it is difficult to see. It is that the community has begun to see something troubling in it, and disagrees about what. Caesar et al. (2018) reconstructed a fingerprint of AMOC weakening from sea-surface temperature patterns and argued that the circulation had reached a millennial minimum. Boers (2021) applied early-warning-signal statistics to the reconstructed record and reported evidence consistent with a system approaching a critical transition. Ditlevsen and Ditlevsen (2023) extended this line of analysis and proposed a probabilistic tipping window centred on the mid-twenty-first century — a claim that received extensive public coverage and equally extensive methodological challenge from within the field (Ben-Yami et al., 2024).⁴ The independent observational record from RAPID, meanwhile, has not shown a monotonic slowdown consistent with the reconstructed multidecadal trend (Ben I Moat et al., 2026; Rahmstorf et al., 2015), leaving the field with a partial and contested empirical picture. This is precisely the kind of system where representational clarity is most needed and least available: a system whose stakes are civilisational, whose evidence is contested at the level of methodology and not merely interpretation, and whose mechanism is invisible to direct observation.

1.4 Greenland as Cognitive Bridge: From the Visible to the Invisible

The AMOC is the system this thesis addresses, but it is not the system the general public sees. When non-specialists are asked to evoke an image of climate change, they rarely nominate a subsurface circulation; they nominate ice (O'Neill & Smith, 2014). The Greenland Ice Sheet is iconic in the specific technical sense used in visual communication studies — it has a documented recurrence in journalism, satellite imagery and public discourse that no oceanic feature can rival. It is one of the small handful of images through which the climate system reaches non-expert cognition at all. The system that is scientifically most consequential, in short, is not the system that is culturally most legible; and the mismatch is not incidental to the argument of this thesis, it is the ground on which the argument is built.

The two systems are, however, physically coupled. Freshwater discharge from the accelerating mass loss of the Greenland Ice Sheet enters the subpolar North Atlantic and reduces surface density; a lower-density surface layer inhibits the deep convective mixing that renews North Atlantic Deep Water; and a weakened deep-water formation weakens the meridional pump that sustains the AMOC (Böning et al., 2016; Caesar et al., 2018; Rahmstorf et al.,

⁴ The methodological reply of Ben-Yami et al. (2024) argues that the confidence intervals in Ditlevsen and Ditlevsen (2023) depend on assumptions about the proxy record — specifically the treatment of the sea-surface temperature reconstruction as a stationary indicator — that are not adequately justified. The dispute is unresolved at the time of writing; the point relevant here is that the disagreement itself demonstrates the representational problem this thesis addresses.

2015).⁵ What the public sees — ice retreating from the terminus of a glacier — is causally connected to what the public does not see: a slowing engine of ocean heat transport that regulates the climate of an entire continent.

This thesis exploits that coupling narratively as well as physically. Greenland enters not as a second case study, whose analytical treatment would double the scope of the work, but as the cognitive entry point through which the invisible mechanism can be approached. It is the piece of the story the reader arrives holding; the argument then leads that reader from the seen to the inferred, from the symptom to the system. This trajectory — from ice to ocean, from what a satellite photograph makes obvious to what only a decade of moored current-meter observations at a single latitude can suggest — is the narrative structure of the experiential prototype developed in Chapter 7, and it is the pedagogical logic that runs beneath every representational decision the thesis will defend.

1.5 On the Interdisciplinary Nature of the Work

The work presented here sits at an intersection that reflects the author’s disciplinary orientation: the meeting point of physical science, computational methods and the perceptual conditions under which non-specialists encounter complex systems. The commitment that runs beneath every methodological choice in the thesis is that these three domains are not adjacent but continuous — that the credibility of a scientific claim, the machinery through which the claim is derived and the form in which it becomes intelligible to a non-specialist reader are not separable problems, and treating them as such is one of the reasons the public understanding of climate risk has been so difficult to advance.

Machine learning is discussed here neither as a predictive technology nor as an automation tool, but in a third and less familiar role: as a *representational substrate* for systems that cannot be observed directly. The proposition is that the latent geometries recovered by unsupervised methods — what a compressed representation preserves and what it discards — can be treated as legitimate objects of design, becoming the coordinate system through which a system too complex for direct intuition is made available to perception. Recent work in the physical sciences has begun to argue for a similar epistemological role for learned representations (Iten et al., 2020), though its extension into public communication is, to the author’s knowledge, largely unattempted. The mission of the thesis is to test that extension in one specific, well-bounded case, with the methodological discipline required to make what is learned transferable beyond it.

⁵ The mechanism is well established in principle but quantitatively uncertain in current attribution studies. Direct RAPID observations at 26° north do not yet resolve the freshwater signal cleanly against interannual variability, and the extent to which the multidecadal reconstructed weakening is attributable to Greenland forcing — rather than to internal variability or to anthropogenic aerosol forcing — remains an active research question (Caesar et al., 2018).

1.6 Purpose, Relevance and Alignment with the Sustainable Development Goals

The relevance of the work proposed here operates at three levels, which the thesis will attempt to substantiate without overclaiming any of them.

The first is academic. The field concerned with the public communication of complex climate systems — what has come to be called *complexity communication* — is methodologically thin. The empirical literature is dominated by preference studies, in which participants are shown alternative visualisations and asked which they find clearer or more engaging (Harold et al., 2016). Comprehension itself, tested against ground-truth understanding of system behaviour, is rarely the object of measurement, and experiential formats — interactive, embodied or sonically encoded — are almost absent from the evaluated evidence base. This thesis contributes not another visualisation nor another survey, but a comparative methodology in which the same underlying data are presented in three representational conditions of increasing experiential richness and evaluated qualitatively for the conceptual understanding they support. To the author’s knowledge, this methodology has not previously been applied to the AMOC or to any system of comparable structural complexity.

The second level of relevance is institutional, and here the alignment with the United Nations Sustainable Development Goals is genuine rather than nominal. The primary anchor is **SDG 13, Climate Action**, and specifically target 13.3, which commits signatories to “*improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning*” (*Transforming Our World: The 2030 Agenda for Sustainable Development* | Department of Economic and Social Affairs, n.d.). Every noun in that sentence — education, awareness, capacity, early warning — names a task the thesis is engaged with. A secondary anchor is **SDG 4, Quality Education**, insofar as the work concerns the pedagogical problem of how conceptually difficult content is transmitted to non-specialists; and a further, more indirect anchor is **SDG 14, Life Below Water**, given that the AMOC is a component of the ocean processes on which marine and coastal systems depend. The alignment claim is that the thesis contributes concrete methodological work to the awareness-raising and capacity-building objectives that these goals name.

The third level is practical, and it concerns the transferability of the methodology beyond its case study. The AMOC is one of an identified family of climate tipping elements — the Greenland and West Antarctic ice sheets, the Amazon rainforest, boreal permafrost, the Indian summer monsoon, the coral reef systems — each of which shares the structural features that make the representational problem hard: invisibility to direct perception, multidecadal to centennial dynamics, non-linear response with hysteresis, and civilisational stakes (Armstrong McKay et al., 2022; Lenton et al., 2008). The framework developed here — machine learning as analytical substrate, experiential representation as communicative output, qualitative evaluation as feedback loop — is designed to be portable. The AMOC is the stress-test on which the framework is validated; it is not the endpoint at which the framework is exhausted. If the argument of the thesis holds, its methodology will be usable, with appropriate domain adaptation, on any climate system whose scientific gravity exceeds its cultural legibility.

The three levels reinforce each other. The academic gap justifies the methodological work; the institutional alignment justifies the choice of case study; and the practical transferability justifies the investment of thesis-scale effort in a single system. Before a society can act on a system, it must first be able to picture it, and this thesis is one contribution to the work of learning how.

1.7 Thesis Statement

This thesis investigates how machine-learning–derived representations of the Atlantic Meridional Overturning Circulation can be designed to improve the understanding of complex climate tipping risk among non-specialist audiences. It proposes that the challenge of communicating systems like the AMOC is not predictive but *representational*: even where the data are available, the system remains cognitively inaccessible. It addresses this challenge by combining machine-learning analysis of ocean transport dynamics with interactive and experiential visualisation, and by evaluating the resulting representations through a small-scale qualitative user study. The contribution offered is not a new oceanographic finding, but an empirical bridge: a transferable methodology for translating the latent structure of an invisible system into a form humans can perceive, and for evaluating, with appropriate epistemic modesty, whether such translation supports understanding. The methodological commitment underlying every choice in the thesis is that machine learning here serves as a *substrate*, not as the deliverable — the deliverable is intelligibility.

1.8 Scope, Boundaries and Honest Limitations

The thesis pursues its argument within deliberate boundaries, and the honest declaration of those boundaries is itself part of the methodological discipline the work claims to embody. Four commitments deserve to be made explicit at the outset.

First, this is not a thesis in oceanography. The primitive equations of ocean circulation (Vallis, 2017), the numerical schemes through which they are solved, and the physical mechanisms through which the AMOC responds to external forcing are treated as inputs, not as objects of contribution. Where these mechanisms are discussed — in Chapter 4 for the theoretical apparatus, and in Chapter 6 for the analytical pipeline — the discussion is at the level of what a well-informed non-oceanographer needs in order to understand the representational problem, not at the level at which the oceanographic community itself debates it.

Second, this is not a thesis in forecasting. A machine-learning approach to the AMOC that would be tempting for its scientific ambition — a recurrent architecture trained to produce short-range probabilistic projections of transport strength with calibrated uncertainty bands — has been considered and explicitly deferred. It is scoped as future work in Chapter 11. The reason for the deferral is not a limitation of the observational record or of computational resources; it is a decision about the argument the thesis is trying to make. Adding a predictive

layer would risk shifting the argumentative centre of gravity from representation — where the thesis claims a contribution — to prediction, where it would be adding, at best, an incremental result to a well-populated literature (Sonnewald et al., 2021).

Third, the empirical study through which the representational framework is evaluated is qualitative and exploratory by design. It engages a small number of participants (eight to twelve) in think-aloud protocols with the three representational conditions, and analyses the resulting transcripts through thematic analysis in the tradition of Braun and Clarke (2006). The thesis does not claim, and its methodology cannot support, statistically powered inference to a wider population. What it claims is what such a study can legitimately claim: an initial mapping of the ways in which non-specialists articulate their understanding of an invisible system under different representational conditions, with the qualitative depth appropriate to a first empirical investigation of an under-studied intervention.

Fourth, the experiential prototype produced in Chapter 7 is a research artefact, not a product. It is designed to be legible to a small number of participants under controlled conditions, not to be deployed to an unbounded public. This distinction matters because a production artefact would require different design choices — around robustness, accessibility, cultural translatability and long-term maintenance — that lie outside the scope of a Master's thesis. The prototype is the vehicle through which the thesis's design claims are made testable; extending it beyond that role would be a research programme of its own.

The pattern in these four boundaries is not defensive. Each of them is a decision, taken consciously, to keep the centre of gravity of the thesis on the specific contribution it is claiming: a methodology for translating an invisible system into an experientially available representation and for evaluating that translation with epistemically honest tools. Scope-freeze is treated here as a methodological principle, not as an apology, and the supervisor's guidance during the development of this work has been that excess of scope, not insufficiency, is the risk against which a thesis of this ambition most needs to be defended.

1.9 Roadmap

The thesis is organised in three parts, each addressing one stage of the detect–translate–evaluate arc that structures the argument.

Part I — Foundations establishes the problem this thesis addresses and the case study through which it is investigated. Chapter 2 formalises the research questions, objectives and hypotheses that structure the work. Chapter 3 reviews the relevant literature across five domains — psychological barriers to climate risk perception, the cognitive science of dynamic systems, empirical evaluation of climate visualisation, embodied and immersive representation, and machine-learning applications to climate science — and identifies the intersection at which this thesis positions itself. Chapter 4 develops the physical and mathematical apparatus needed to treat the AMOC as a measurable and analysable object.

Part II — Translation documents the constructive work of the thesis. Chapter 5 specifies the three-stage methodology — analytical, design-based and empirical — through which the

argument is made testable. Chapter 6 presents the analytical pipeline: the dimensionality-reduction, anomaly-detection and early-warning analyses through which machine learning recovers the latent structure of the observational record, and reports the results that motivate the representational choices that follow. Chapter 7 designs and documents the three representational conditions — conventional static, interactive dynamic and machine-learning-driven experiential — through which that latent structure is translated for non-specialist audiences.

Part III — Evaluation and Reflection closes the argumentative loop. Chapter 8 specifies the qualitative user study through which the three conditions are compared. Chapter 9 reports its findings. Chapter 10 interprets those findings in conversation with the literature reviewed in Part I and with the analytical results of Part II, and Chapter 11 concludes with reflections on limitations, transferability to other invisible climate systems and future work.

Chapter 2

Research Questions, Objectives and Hypotheses

2.1 Research Framework

The argument developed in Chapter 1 identified a representational problem: complex climate tipping systems remain cognitively inaccessible to non-specialist audiences even when the underlying data are open, abundant and technically legible. This chapter formalises the research programme through which that problem is addressed. The programme is organised around a single central research question and three subordinate sub-questions, each of which investigates the central question through a different epistemic mode. This structure is not a matter of layout: it reflects the actual architecture of the work. The three sub-questions are not three parallel investigations that happen to share a topic. They are three complementary observations of the same phenomenon — the way in which representation mediates understanding — taken with instruments of different kinds.

The framework that organises them is the tripartite arc of *detect, translate, and evaluate*. This arc is not novel: it belongs to the family of design-based research methodologies developed in the learning sciences over the past two decades (Anderson & Shattuck, 2012; Barab & Squire, 2004), which treat design not as an ancillary activity to research but as a mode of enquiry with its own epistemic commitments. In this tradition, an artefact is designed in response to a theoretically grounded problem, its design is documented as a form of knowledge production, and its consequences are evaluated empirically in the context for which it was designed. The three phases feed one another: what is detected constrains what can be translated; what is translated becomes the object of evaluation; what is evaluated returns to inform the detection and design of subsequent iterations. The three sub-questions of this thesis correspond to these three phases, and the central research question is the question that all three sub-questions are attempting, from different angles, to answer.

2.2 Research Questions

The central research question orients the thesis as a whole:

RQ. *How can machine-learning-derived representations of AMOC instability be designed to improve public understanding of complex climate tipping risk?*

Every clause of this question carries argumentative weight. The verb *designed* places the enquiry within the design-based tradition rather than the predictive one; the beneficiary *public understanding* names an outcome that is neither scientific prediction nor visual aesthetics but the mediating cognitive process between them; the domain *complex climate tipping risk* identifies a class of phenomena of which the AMOC is one representative case. The question does not ask whether machine learning can detect an AMOC tipping point, a question already contested within oceanography (Ben-Yami et al., 2024; Ditlevsen & Ditlevsen, 2023). It asks whether the latent structure that machine learning recovers from the observational record can be made available to non-specialist perception in ways that support the understanding of a system too complex for direct intuition.

Three subordinate research questions articulate the central enquiry through the detect–translate–evaluate arc.

RQ1 (analytical). *What latent structural patterns and dynamical signatures of AMOC variability can machine learning methods recover from the RAPID observational record and the Copernicus reanalysis ensemble, and how do these representations compare with what classical statistical approaches capture?*

This sub-question is descriptive and exploratory in its epistemic character. It does not commit the thesis to finding a novel oceanographic result. It commits the thesis to investigating, honestly and comparatively, what is in the data — and what is not — when the data are analysed with methods drawn from both dimensionality-reduction traditions and classical dynamical statistics. The value of the analytical phase, as Chapter 6 will demonstrate, lies in what it authorises the next phase to attempt.

RQ2 (design-based). *How can the latent structures recovered in RQ1 be translated into interactive and experiential representations that encode the AMOC’s dynamics, invisibility and tipping-point risk in perceptually meaningful form?*

This sub-question is generative rather than descriptive. Its epistemic character is that of design research: the question does not have a single correct answer to be discovered, but a class of well-justified answers to be constructed, documented and defended. Chapter 7 answers RQ2 by producing three representational conditions of increasing experiential richness, each grounded in a documented design grammar that maps machine-learning-derived features onto perceptual dimensions in ways consistent with established principles of embodied cognition and pre-attentive visual processing.

RQ3 (empirical). *To what extent do experiential representations of AMOC dynamics, compared with conventional static visualisation, support the ways in which non-specialist audiences articulate their conceptual understanding of complex system behaviour?*

This sub-question is comparative and empirical, but qualitative and exploratory in scale. It does not claim to test the population-level effectiveness of experiential representations, which

would require a study design outside the scope of this thesis. It claims something more modest and more scientifically honest: to observe closely, through the think-aloud methodology, the ways in which a small number of non-specialist participants articulate their understanding of the AMOC when they encounter it in three progressively richer representational forms. The evidence such a study can bear is qualitative depth, not statistical inference, and the phrasing of the question reflects that commitment explicitly.

Taken together, the three sub-questions constitute the operational articulation of the central question. RQ1 establishes what there is to represent; RQ2 constructs the representations; RQ3 investigates whether the representations do the work the central question asks of them. No single sub-question can answer the central question alone. Their combination is the answer.

2.3 Objectives

Each sub-question translates into a set of research objectives, formulated with active verbs and specified outputs. The objectives are the operational commitments the thesis makes: what it undertakes to produce, and how those products will be recognisable in the chapters that follow. Ten objectives across the three sub-questions define the concrete work of the thesis.

The analytical objectives derived from RQ1 organise the work of Chapter 6. **O1.1** decomposes the vertical structure of AMOC variability from the RAPID stream-function record using principal component analysis and identifies the dominant modes of circulation behaviour. Its output is a set of interpretable latent components, quantified by explained variance and characterised through comparison against the physical modes recognised in the oceanographic literature (Buckley & Marshall, 2016). **O1.2** compares the linear latent representation obtained via PCA against a non-linear autoencoder architecture at matched latent dimensionality, in order to assess whether the AMOC vertical manifold exhibits non-linear structure exploitable for representation. Its output is a comparative reconstruction analysis whose interpretation, developed in Chapter 6, has direct implications for the coordinate system on which the representational work of Chapter 7 is built. **O1.3** applies classical Early Warning Signal methodology and unsupervised anomaly detection to the RAPID record, and characterises their convergence and divergence with respect to known dynamical events. Its output is a rigorous statistical analysis with corrected significance testing, in which methodological honesty about statistical power is treated as itself a form of scientific contribution. **O1.4** specifies, and explicitly defers to future work, a probabilistic short-range forecasting layer (an LSTM-based architecture) for the AMOC transport time series. Its output is a scoped future-work specification with documented methodological rationale, presented in Chapter 11. The choice to specify the forecasting layer without implementing it is a deliberate act of scope discipline, addressed further in §2.6.

The design objectives derived from RQ2 organise the work of Chapter 7. **O2.1** designs three representational conditions of increasing experiential richness: a conventional static condition (C1), an interactive dynamic condition (C2), and a machine-learning-driven experiential condition (C3). Each condition encodes the same underlying data through a different perceptual

mode, ensuring that any differences observed in the evaluation phase are attributable to representational form rather than to informational content. Its output is three working prototype conditions accompanied by documented design rationales. **O2.2** develops a design grammar that maps machine-learning-derived features — latent position, anomaly score, ensemble uncertainty — onto perceptual dimensions such as colour, motion, density and sound, grounded in principles of embodied cognition and pre-attentive processing (*The Sonification Handbook* | Edited by Hermann, Hunt, Neuhoff, n.d.; Ware, 2021). Its output is a documented, transferable design grammar that the thesis intends to be reusable for representational work on other invisible climate systems. **O2.3** encodes the boundary between the historical observational record and the region of probabilistic uncertainty as an experiential perceptual quality — rather than as a numerical annotation, an error bar or a shaded envelope. Its output is a prototypical implementation of experiential uncertainty encoding, defended in Chapter 7 as a design contribution to the broader problem of uncertainty communication.

The empirical objectives derived from RQ3 organise the work of Chapters 8 and 9. **O3.1** designs a qualitative evaluation instrument comprising a think-aloud protocol, a semi-structured post-task interview and a thematic coding scheme, appropriate to a small-N exploratory study (Braun & Clarke, 2006). Its output is a reusable evaluation instrument together with ethics documentation in Appendix 11.1 and study protocols in Appendix 11.1. **O3.2** conducts an exploratory user study with 8 to 12 participants of mixed technical background, exposing each participant to the three representational conditions and recording their articulated understanding of the AMOC's behaviour under each. Its output is a set of anonymised transcripts and a thematic analysis reported in Chapter 9. **O3.3** assesses whether the machine-learning-derived perceptual encodings introduced in condition C3 — latent-structure transitions and ensemble uncertainty as felt quality — are correctly interpreted by non-specialist participants, or whether they add representational noise. Its output is an honest reporting, in Chapter 10, of the communicative effectiveness of the machine-learning-as-perceptual-layer approach, defended regardless of whether the finding is positive or negative.

The relationship between objectives, hypotheses and chapters is summarised in the alignment matrix presented in §2.5. Each objective corresponds to a chapter in which its output is realised, and to a claim (hypothesis, design proposition or empirical expectation) formulated in §2.4. This traceability between what the thesis undertakes and where it delivers is the structural mechanism through which the thesis makes itself accountable to its own commitments.

2.4 Hypotheses and Design Propositions

The three sub-questions differ in epistemic character, and the claims this thesis advances are framed accordingly. RQ1 is analytical and exploratory, and admits directional hypotheses about what the machine-learning analysis will reveal in the observational record. RQ2 is design-based: its claims take the form of *design propositions*, which are realised and assessed rather than confirmed or refuted. RQ3 is empirical, but qualitative and small in scale; its claims are therefore stated as *expectations* that orient the analysis rather than as hypotheses subjected

to statistical test. Stating this distinction explicitly is itself a methodological commitment: a directional hypothesis, a design proposition and an exploratory expectation are different kinds of claim, and conflating them would misrepresent the evidence each can bear (*Conducting Educational Design Research*, n.d.; Cronbach, 1975).

Two *umbrella hypotheses* frame the enquiry as a whole. They are not tested through a single instrument but through the coordinated evidence produced by all three sub-questions, and their fate is discussed in Chapter 10.

H-A (the problem). *Conventional representations of complex climate tipping systems — static, graphic and quantitatively annotated — are perceptually suboptimal for non-specialist audiences, because they treat systemic complexity as an object to be read rather than as a dynamic to be inhabited.*

H-B (the response). *Machine-learning-derived representations, translated into interactive and experiential forms, can support the conceptual understanding, engagement and retention of complex system behaviour among non-specialist audiences more effectively than the conventional format, not by replacing quantitative analysis but by rendering its structure perceptually available.*

Analytical hypotheses (RQ1)

Three directional hypotheses guide the analytical phase. Their empirical fate is reported in Chapter 6 and interpreted in Chapter 10.

H1. *A small number of linear modes accounts for the dominant share of variance in the AMOC vertical profiles, reflecting a circulation regime governed by a few large-scale modes rather than by high-dimensional structure (Buckley & Marshall, 2016; Hannachi et al., 2007).*

H2. *A non-linear autoencoder does not materially outperform linear PCA at equal latent dimensionality, indicating that the AMOC vertical manifold is essentially flat and that interpretable linear coordinates lose little relative to a more expressive model (Baldi & Hornik, 1989).*

H3. *Classical Early Warning Signals — rising lag-1 autocorrelation and rising variance — show a statistically significant monotonic trend across the RAPID record, once the effective sample size is properly accounted for through correction for autocorrelation of overlapping windows (Dakos et al., 2012; Scheffer et al., 2009).*

Design propositions (RQ2)

The design phase advances propositions rather than hypotheses, because a representation is realised, defended and assessed against design criteria rather than falsified against ground truth. The realisation of these propositions is documented in Chapter 7.

DP2.1. *Machine-learning-derived features — latent position, anomaly score, ensemble uncertainty — can be mapped onto perceptual dimensions such as colour, motion, density and sound in a manner consistent with established principles of pre-attentive processing and embodied cognition (Lakoff, n.d.; Ware, 2021).*

DP2.2. *The boundary between the historical observational record and the region of probabilistic projection can be encoded as a felt perceptual transition rather than as a numerical annotation, rendering uncertainty an experiential quality rather than a bracketed error.*

Empirical expectations (RQ3)

Because the user study is qualitative, exploratory and small in scale, its guiding claims are stated as expectations that orient the thematic analysis, not as hypotheses subjected to inferential test. The empirical study through which these expectations are engaged is designed in Chapter 8 and reported in Chapter 9.

E3.1. *Increasing experiential richness across the three representational conditions is associated with differences in how participants articulate their understanding of the system’s behaviour, rather than with a measurable gain in comprehension scores.*

E3.2. *The perceptual encoding of uncertainty in condition C3 is interpreted by non-specialist participants in ways that the think-aloud protocol can surface, whether or not the interpretation matches the intended reading.*

The empirical fate of H1–H3 is reported in Chapter 6 and revisited in Chapter 10; DP2.1 and DP2.2 are realised in Chapter 7 and defended there; E3.1 and E3.2 frame the interpretive analysis of Chapters 9 and 10. This structural correspondence between claim and chapter is the mechanism through which the thesis makes itself accountable to its own commitments, and it is made visible at a glance in the alignment matrix presented in §2.5.

2.5 Alignment Matrix

The alignment matrix (Table 2.1) synthesises the correspondence between the three sub-questions and the operational structure of the thesis. Each column represents one sub-question and one phase of the detect–translate–evaluate arc; each row identifies a category of correspondence — the epistemic character of the contribution, the chapters in which it is delivered, the objectives that operationalise it, the claims (hypotheses, design propositions or empirical expectations) it advances, the material with which it works, the output it produces, its alignment with the Sustainable Development Goals and the explicit scope discipline that keeps it defensible in the time available. The matrix is intended to be readable at a glance, and it functions as a promissory contract between the thesis and its reader: for every commitment made in this chapter, a locatable delivery in a specific later chapter.

Table 2.1: Alignment between the three sub-questions and the operational structure of the thesis.

	RQ1 — Detect	RQ2 — Translate	RQ3 — Evaluate
Epistemic character	Analytical, comparative and exploratory	Design-based and generative	Empirical and qualitative
Chapters	Ch. 4, Ch. 6	Ch. 5, Ch. 7	Ch. 5, Ch. 8, Ch. 9
Objectives	O1.1–O1.4	O2.1–O2.3	O3.1–O3.3
Claims	H1, H2, H3	DP2.1, DP2.2	E3.1, E3.2
Material	RAPID observations, Copernicus reanalysis ensemble	Latent structures recovered in RQ1	Prototypes produced in RQ2, participants
Output	Latent structure analysis; EWS null result; anomaly characterisation	Three representational conditions; documented design grammar	Anonymised transcripts; thematic analysis; cross-condition comparison
SDG anchoring	13 (Climate Action), 14 (Life Below Water)	4 (Quality Education), 13	4, 13
Scope discipline	LSTM forecasting deferred as future work	Research prototype, not production artefact	Exploratory qualitative, not statistically powered

Reading the matrix vertically clarifies the three coherent programmes of work the thesis undertakes: an analytical programme grounded in the observational record, a design programme grounded in the analytical outputs and an empirical programme grounded in the design outputs. Reading the matrix horizontally makes explicit the mechanism through which each phase feeds the next: the material of RQ2 is the output of RQ1, and the material of RQ3 is the output of RQ2. Neither reading is complete on its own; the alignment matrix is legible only through both.

2.6 Feasibility and Scope Discipline

The programme of work formalised in this chapter is ambitious in its combination but bounded in its individual commitments, and this section makes the reasoning behind that balance explicit. Three considerations govern the feasibility of the thesis as a whole: the availability of the material infrastructure required to execute each phase, the temporal budget available for its completion and the discipline of scope through which each phase has been protected from overextension.

The material infrastructure is largely in place at the moment of writing. The analytical phase, corresponding to RQ1, is substantially complete: the pipeline over the RAPID observational record and the Copernicus reanalysis ensemble has been developed across five sequentially numbered Jupyter notebooks, and its principal quantitative results — the linear latent decomposition, the autoencoder comparison, the corrected Early Warning Signals analysis and the anomaly characterisation — are reported in Chapter 6. The analytical ob-

jectives O1.1 through O1.3 have therefore already been operationalised; O1.4, concerning the deferred forecasting layer, is by design a specification exercise rather than an implementation one. The empirical infrastructure for RQ3 is comparatively lightweight: eight to twelve participants recruited from a mixed technical background, a think-aloud protocol implemented in individual sessions of forty to sixty minutes each, transcription supported by open-source speech-to-text tooling and thematic analysis conducted in NVivo or a comparable qualitative-analysis environment. The recruitment target is realistic for the timeframe, and the exploratory qualitative framing — explicitly endorsed in supervisory guidance — releases the study from the sample-size requirements of statistically powered designs (Braun & Clarke, 2006; Smith & McGannon, 2018). The infrastructure required by RQ2, the design phase, is the most substantial and is discussed in Chapters 5 and 7.

The temporal budget is the principal binding constraint. The thesis is scheduled for completion by the end of August 2026, which allocates approximately nine to ten weeks from the writing of this chapter to the final submission. The distribution of work across that budget is planned as follows: the completion of the theoretical chapters (3 and 4) and the drafting of the methodology (Chapter 5) in the first three weeks; the design and implementation of the three representational conditions (Chapter 7) across four weeks, with parallel drafting of the analytical results chapter (Chapter 6) which reports work already conducted; the user study (Chapters 8 and 9) across two weeks; and the discussion, conclusion and final revisions across the last week. This allocation deliberately front-loads the theoretical and analytical writing, so that the technically riskiest phase — the implementation of condition C3 in Chapter 7 — is undertaken with the surrounding text already stable.

The discipline of scope is treated in this thesis not as a concession but as a methodological principle, and it operates at three distinct levels of the programme. At the level of the analytical phase, the forecasting layer has been specified but deferred, on the grounds that its addition would shift the argumentative centre of gravity from representation — where the thesis claims a contribution — towards prediction, where it would be adding only an incremental result to a well-populated literature (Sonnewald et al., 2021). At the level of the design phase, the experiential prototype is constrained to be a research artefact rather than a production system: it is designed to be legible to a small number of participants under controlled conditions, not to be deployed to an unbounded public audience whose diverse accessibility, robustness and cultural-translatability requirements would define a research programme of their own. At the level of the empirical phase, the user study is qualitative and exploratory: it does not claim, and its design cannot support, statistical inference to a wider population. These three constraints are not weaknesses admitted defensively; they are decisions taken consciously, defended in Chapter 1 in the discussion of scope (§1.8), and mirrored in the sub-question formulations of §2.2.

Two flexibility mechanisms are embedded in the plan. If the user study slips beyond its allocated window, the design and analytical chapters (Chapters 7 and 6) can absorb the additional writing time without compromising the analytical claims. If the technical implementation of condition C3 encounters an obstacle that cannot be resolved within the four allocated weeks, the design grammar remains a documented contribution in its own right

(O2.2), and Chapter 7 can defend the thesis's design claims with the two simpler conditions and the specification of the third. These are not fallback positions imagined against failure; they are the mechanism by which the thesis has been designed to remain scientifically coherent under realistic variations in its execution.

The ambition of this thesis is thus limited by design, not by circumstance. Each of the objectives formulated in §2.3 is scoped to be achievable within the temporal and material budget available, and the alignment between claim, method and chapter documented in §2.5 is the structural mechanism through which the coherence of the whole is protected. The remaining chapters are the delivery of the commitments this one has made.

Chapter 3

Literature and Market Review

The literature reviewed in this chapter organises around five strands that together define the intersection at which this thesis positions itself: the psychology of climate risk perception, the cognitive science of dynamic and complex systems, the empirical evaluation of climate visualisation, the strands of data humanism, embodied cognition and immersive analytics, and the applications of machine learning to Earth-system science. Each strand is developed as a compact argument rather than as an inventory of publications; the sixth and final section synthesises them into the specific gap the thesis addresses.

3.1 Psychological Barriers to Climate Risk Perception

For much of the late twentieth century, the dominant model of climate communication rested on what became known as the information-deficit hypothesis: the assumption that public inaction on climate change reflected a shortage of accurate scientific information, and that closing the knowledge gap would close the action gap. Two decades of empirical research have not been kind to that assumption. Kellstedt et al. (2008) showed, in a representative American survey, that respondents who scored higher on climate knowledge expressed *lower* personal concern about the issue and *lower* sense of responsibility for addressing it, an inverse relationship that persisted after controlling for political affiliation and demographic variables. Kahan (2015), extending this line of work, demonstrated that scientific literacy interacted with cultural worldview to *polarise* rather than converge climate attitudes, so that the most scientifically literate members of opposing cultural groups disagreed with each other more sharply than their less-literate counterparts. The uncomfortable finding of this literature is not that information does not matter but that its effects are mediated by cognitive and cultural machinery which the standard formats of climate communication were not designed to address. What has failed is not the messenger's accuracy but the messenger's understanding of who is listening and how.

The most influential systematic account of this failure remains Stoknes (2015), whose typology of five psychological barriers has been widely adopted in the science-communication literature and has structured much of the applied research that followed. The five barriers, which Stoknes labels the *five Ds*, are *distance* (climate change is perceived as spatially, temporally and socially remote from everyday experience), *doom* (catastrophist framing triggers avoidance rather than engagement), *dissonance* (the gap between climate belief and daily behaviour is resolved by adjusting the belief rather than the behaviour), *denial* (self-protective

disengagement from information experienced as threatening) and *identity* (climate positions function as markers of group belonging, making the acceptance of new information contingent on its compatibility with prior identification). The five barriers do not operate independently; Van Der Linden (2015) document their interlocking action empirically and argue that effective communication requires addressing them jointly rather than sequentially. The finer-grained segmentation offered by *Global Warming's Six Americas, September 2021* (n.d.), in the long-running *Six Americas* study of the Yale Program on Climate Change Communication, shows that publics stratify into psychologically coherent audiences whose barriers activate differently, so that no single communicative strategy can serve all of them at once. The tempting inference — that better graphs might solve this — is not supported by the evidence the following section examines.

The reason conventional visualisation has failed to dissolve the barriers identified by Stoknes and his successors is structural, not aesthetic. Creative (n.d.), drawing on empirical work conducted by the Climate Outreach organisation in the United Kingdom, showed that the imagery most frequently used to communicate climate change — polar bears, smokestacks, calving glaciers, temperature curves — either reinforces the very psychological distance the communicators hope to close or activates the doom response that suppresses engagement. Nicholson-Cole (2005), in what remains a foundational study of the perceptual reception of climate visuals, demonstrated that participants presented with standard scientific climate imagery reported a paradoxical combination of concern and detachment: the imagery made the phenomenon feel real, but it also made it feel finished, already someone else's problem, cognitively closed. The recent review of Ojala (2023) synthesises a decade of subsequent work on affective responses to climate communication and concludes that formats which do not offer the reader an active role in relation to the information consistently underperform formats that do. Adding more precision to a line graph does not open such a role; the graph's implicit rhetoric of *here is the fact* leaves the reader with nowhere to stand.

The direction in which the more recent literature has begun to point is towards representational forms that treat the reader as an active participant in the construction of meaning rather than as the passive recipient of an authoritative image (Marlon et al., 2019; Ojala, 2023). This shift has taken several forms: narrative-based communication that embeds climate information in stories with which readers can identify (Morris et al., 2019), visualisation practices that explicitly invite exploration and interpretation rather than mere reading (Creative, n.d.) and, at the more experimental end, representational practices grounded in embodied and interactive cognition to which this thesis contributes, and which the fourth section of this chapter (§3.4) develops in detail. The psychological literature identifies the problem to which the representational literature has yet to find an adequate answer, and the case made in this thesis is that machine learning, treated not as a predictive instrument but as a substrate for representation, offers one path through which that answer might be pursued.

3.2 The Cognitive Science of Dynamic and Complex Systems

The psychological literature reviewed in the previous section identified the cultural and affective barriers through which climate information is filtered before it reaches interpretation. The cognitive-science literature reviewed here identifies a barrier of a different kind, prior to filtering and independent of it: the structural limits of the human mind in reasoning about systems whose behaviour is governed by accumulation, feedback and delay. The claim developed in this section is that these limits are not defects of education which better teaching could remedy but stable features of human cognitive architecture, and that any representational strategy which ignores them will fail regardless of how psychologically well-tuned it might be.

The most systematic body of evidence for this claim comes from the work of Sterman (2008), whose two decades of experimental studies have consistently shown that even mathematically literate adults — graduate students at MIT, engineering professionals, participants with formal training in calculus and dynamics — make systematic errors when asked to reason about elementary stock-and-flow systems. The experimental design is deceptively simple. Participants were shown a scenario in which atmospheric CO₂ rises gradually and stabilizes at 400 ppmv by 2100 (Figure ??), together with the historical trajectory of anthropogenic emissions and the current rate of net removal by oceans and biomass (Figure ??). They were then asked to sketch the emissions path consistent with stabilization. As Figure ?? illustrates, 84% of respondents produced trajectories that violate mass balance: rather than reasoning about the relationship between inflow and outflow, they matched the shape of the output to the shape of the input, keeping emissions well above removal while expecting the stock to stabilize.

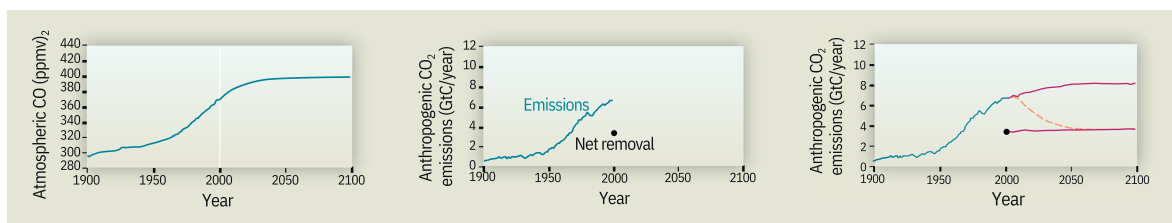


Figure 3.1: The climate stabilization task. (a) The scenario presented to participants: atmospheric CO₂ concentration rises gradually to 400 ppmv and stabilizes by 2100. (b) The stock–flow information provided: anthropogenic CO₂ emissions from 1900 to 2000 (the inflow) and current net removal by oceans and biomass (the outflow), with emissions roughly double net removal. (c) A typical response: participants erroneously sketch future emissions that match the shape of the CO₂ trajectory (pattern-matching heuristic), while mass balance requires emissions to fall until they equal net removal (dashed line). Reprinted from Sterman (2008).

Cronin et al. (2009) demonstrated the finding in its starkest form: participants shown a graph of the inflow and outflow of some quantity into a stock were asked to sketch the trajectory of the stock itself, a task equivalent to visually integrating the difference between two curves. Even in scenarios simplified to the point of triviality, the majority produced trajectories that confused rates with levels, mistook peaks in the flow for peaks in the stock

and treated the dynamic system as if it were static. The pattern was not distributed randomly across the participant pool; it appeared in the responses of participants who had scored highly on prior tests of mathematical competence and who had, in principle, the technical apparatus to solve the problem correctly. What the study documented was not a failure of arithmetic but a failure of representation — the mind, given only a graph, does not spontaneously construct the dynamic scenario the graph is meant to depict. The implication for climate communication is direct and uncomfortable: presenting CO₂ emissions and CO₂ concentrations as parallel curves does not communicate that the second is the integral of the first, and readers of the graph will systematically underestimate the inertia of the system (Stermann, 2008).

A second, converging line of evidence comes from cognitive-load theory as developed by Sweller (1988) and extended over three decades of subsequent research (Sweller et al., 2019). The theory rests on a well-documented empirical fact: working memory, the mental resource in which active reasoning occurs, has a strictly limited capacity, and any representational format which requires the reader to hold multiple relationships in working memory simultaneously will produce comprehension errors that scale with the number of relationships. Static visualisations of dynamic systems — an ensemble spaghetti plot, a set of stacked time series, an isocontour map of projected change — ask the reader to mentally animate what the visualisation cannot animate directly, to hold in mind the transitions between frames the visualisation cannot show, to construct an internal simulation of a process the graph depicts only in its final state. The cognitive load imposed by this reconstruction is high, and it is imposed *in addition* to the load of understanding the domain itself. The failures of comprehension that result are not deficiencies of the reader; they are predictable consequences of asking working memory to do work for which it does not have capacity. Ware (2021), integrating cognitive-load theory with perceptual research on visual attention, argues that the design of any communicative visualisation must minimise this internal-simulation demand by making the dynamic structure of the system perceptually available in the representation itself.

The third strand of evidence gestures towards the direction in which the following sections will develop. Tversky (2019), in a synthesis of four decades of research on spatial and mental representations, argues that human cognition of complex processes is fundamentally spatial in character: the abstract relationships that constitute a dynamic system are understood through analogy to bodily, spatial and kinaesthetic schemas that the mind already inhabits. This finding is more than a fact about mental imagery; it identifies the cognitive substrate through which any representation of a complex system must be received in order to be understood at all. If systems like the AMOC are structurally dynamic, and human reasoning about dynamic systems is spatially embodied, then the representational strategies most likely to succeed are those which enrol the reader's spatial and kinaesthetic cognition in the reconstruction of the dynamic. This proposition, which the next section develops through the literature on embodied cognition and data physicalisation, is not offered here as a conclusion but as the hypothesis under which the remainder of this chapter is organised.

3.3 Empirical Evaluation of Climate Visualisation

The two previous sections established, on the one hand, the psychological and cultural barriers that filter climate information before it reaches interpretation and, on the other hand, the structural limits of cognition that shape what any representation can hope to communicate. This section turns from theory to evidence and asks what has actually been measured about the reception of climate visualisations, and where the resulting empirical literature is thin. The picture that emerges is one of a field methodologically asymmetric: rich in studies of what audiences *prefer* among alternative visualisations and sparse in studies of what audiences actually *understand* when they view them. This asymmetry, more than any single deficit of aesthetics or interactivity, defines the gap this thesis positions itself to address.

The most systematic mapping of the state of the field is offered by Harold et al. (2016), whose review examined the literature on climate-data visualisation against what is known from cognitive and perceptual science about non-expert visual processing. Their central finding is that the standard formats used to communicate climate change — line graphs of global mean temperature, choropleth maps of projected change, ensemble spaghetti plots, tornado diagrams of sensitivity — are mismatched to the cognitive resources of the audiences they are meant to inform, and that the mismatch is documented not only in laboratory studies with student participants but in the interpretive errors made by policy audiences with substantial technical training. The finding is corroborated by Lang (2014), whose study of how policy actors interpret IPCC visualisations demonstrated that even trained readers systematically underestimate uncertainty ranges depicted through shaded envelopes and misread ensemble spread as evidence of disagreement rather than as a structural feature of the projection. The specifically visual literature converges with the psychological literature reviewed in §3.1: what has failed is not the accuracy of the visualisations but their compatibility with the perceptual machinery through which they are received. Retchless and Brewer (2016), in a controlled study of alternative encodings for climate uncertainty on choropleth maps, showed that the encodings most commonly used in scientific practice — bivariate colour schemes and stippled overlays — produced the poorest comprehension among non-specialist participants, while encodings borrowed from cartographic design traditions performed measurably better. The finding is significant not because it identifies a superior encoding but because it establishes that alternatives exist and that they can be evaluated empirically, a claim which much of the visualisation-design literature makes only in principle.

The more troubling structural finding of the field, however, is what Kause et al. (2021) identifies as the persistent gap between preference and comprehension. Their study, one of the few that measured both variables in the same participants, demonstrated that the visualisations participants *preferred* — those they rated as clearest, most engaging and most trustworthy — were not the same visualisations that produced the highest measured comprehension of the underlying phenomenon. Participants preferred formats that felt familiar and confident; they understood formats that were structurally aligned with the dynamic being depicted, whether or not they preferred them. A subsequent extension of the work (Kause et al., 2019) documented the same divergence for the specific case of climate uncertainty, showing that

intuitive-seeming icon arrays and confidence bars produced overconfident interpretation while less intuitive but structurally faithful frequency representations produced calibrated interpretation. The methodological consequence is direct and rarely stated: the dominant tradition of preference-based evaluation in the visualisation literature is measuring the wrong variable. Preference is not comprehension, and optimising representations for preference alone risks constructing what Sheppard (2015) has called *consoling visualisations* — images that are received warmly and understood poorly.

The gap in the empirical literature that this thesis addresses can now be stated with precision. Comprehension studies of climate visualisation exist but are outnumbered by preference studies by roughly an order of magnitude, and among the comprehension studies that exist almost none has evaluated experiential or embodied representational formats against conventional graphic ones (Harold et al., 2016; Sheppard, 2015). Where comparative studies of alternative formats have been conducted, they have almost always relied on quantitative comprehension tests with population-level inference as their epistemic goal, an approach appropriate for well-defined comprehension outcomes but poorly suited to the exploratory investigation of how non-specialists articulate their understanding of a system for the first time. The methodological space this thesis occupies — a qualitative, exploratory comparison of three representational conditions using think-aloud protocols and thematic analysis — is not a replacement for those quantitative traditions but a complement to them, appropriate to a stage of research in which the phenomenon of interest is the *articulation of understanding* itself, and where the depth of interpretive engagement matters more than the breadth of statistical inference (Braun & Clarke, 2006; Smith & McGannon, 2018). The evidence base into which this thesis reports is the one identified by the empirical literature as most thinly populated, and it is populated using the methodology the empirical literature identifies as best fit for the question being asked.

3.4 Data Humanism, Embodied Cognition and Immersive Analytics

The three previous sections defined the problem this thesis addresses through the successive lenses of psychology, cognitive science and empirical visualisation research. This section turns to the traditions in which the problem's response has begun to take shape. Three related lines of work — the philosophical account of embodied cognition, the practice of data humanism and physicalisation and the emerging field of immersive analytics — have converged over the past two decades on a proposition that runs against the mainstream of scientific visualisation: that the intelligibility of complex information depends on the reader's active bodily and perceptual engagement with it, and that representational forms which do not enrol the body and the senses systematically underperform forms that do. Each of the three traditions arrives at this proposition from a different direction, and their convergence provides the theoretical scaffolding on which the design work of Chapter 7 is built.

The philosophical grounding of the position was articulated most systematically by Lakoff

(n.d.), whose work on embodied cognition argues that the abstract concepts through which humans reason about complex systems are not disembodied logical structures but extensions of the bodily and spatial experience that human cognition originally evolved to handle. Concepts such as *acceleration*, *feedback*, *stability* and *tipping* are understood, on this view, through the extension of kinaesthetic schemas that the mind has already inhabited — the sense of a body speeding up, the sense of a movement returning to itself, the sense of a footing that holds, the sense of a footing that gives way. Kirsh (2013) provided empirical support for this position in a series of experimental studies showing that participants who were allowed to gesture and move while reasoning about complex processes performed measurably better than participants restricted to inspection of static representations. The finding held across problem types — mathematical, mechanical and dynamical — and its consistency has since supported a growing body of design research arguing that representations of complex systems should be understood not as pictures to be viewed but as environments to be inhabited (Baur et al., n.d.; Tversky, 2019). The implication for climate visualisation is not that graphs should be replaced by simulations but that any visualisation of a system whose reader is expected to reason about dynamic behaviour must offer the reader a form of embodied engagement with the dynamic, whether kinaesthetic, exploratory or perceptual, that inert graphic formats do not provide.

The design tradition that has developed most concretely from this proposition is the practice of *data physicalisation*, defined by Jansen et al. (2015) as the construction of physical artefacts whose geometry and material properties encode information for exploration through touch, sight and sometimes sound. The tradition emerged partly in reaction to the perceived limitations of purely visual representation and partly from the design-research community's engagement with tangible interaction (Ishii & Ullmer, 1997), and it has since produced both craft objects intended for public engagement and rigorous experimental studies comparing physicalisations against equivalent screen-based visualisations. “[Cover Art]” (2009) showed, in one of the earliest such comparisons, that participants working with data physicalisations recalled the represented information with greater accuracy and retained it longer than participants working with equivalent bar charts, a finding that has since been extended across data types and modalities (Jansen et al., 2015). Parallel to this technical tradition, the practice of *data humanism* associated with the work of Giorgia Lupi (PrintMag, 2017) and colleagues at Accurat, Pentagram and Domestic Data Streamers has argued that scientific data communication has been dominated by an aesthetic of dispassionate abstraction that treats the reader as a rational information processor rather than as an embodied and emotional being. The data-humanist alternative — often executed as hand-drawn or richly illustrated visualisations that foreground authorial voice, imperfection and narrative — has been received with mixed reactions in the scientific-visualisation literature, but the empirical case for its central claim is stronger than its stylistic surface suggests. Studies of narrative visualisation (Segel & Heer, 2010) and of visual explanations grounded in personal experience (Morris et al., 2019) document consistently that representations which acknowledge the reader as a participant produce measurably deeper engagement than representations that maintain the fiction of a view from nowhere.

The third convergent tradition, more recent and more technically ambitious, is the field of *immersive analytics* as consolidated in the survey volume by Marriott et al. (2018). The

field takes as its premise that the immersive display technologies now becoming accessible — head-mounted virtual reality, augmented reality, large-format wall displays and spatially-aware three-dimensional interfaces — offer representational affordances that the two-dimensional screen cannot match, and that these affordances are particularly well-suited to data whose structure is inherently spatial, dynamic or multidimensional. Immersive analytics is distinct from earlier scientific visualisation in its explicit orientation towards *analysis by exploration*: rather than presenting a finished view of a dataset, the immersive representation invites the analyst to move through the data, to change viewpoint, to select and to compare, in ways that leverage the same embodied cognitive resources identified by Lakoff and Kirsh (Chandler et al., 2015). Empirical evaluations of immersive analytics for scientific data are still limited in number, and the strongest current evidence concerns tasks involving three-dimensional spatial structure and temporal dynamics rather than more abstract representations of complex systems. But the theoretical alignment between immersive analytics and the cognitive-science literature reviewed in §3.2 is direct, and the practical accessibility of the underlying technologies — browser-based WebGL for interactive rendering, the Web Audio API for real-time sonification — has now brought immersive-analytic design within reach of individual research projects at the scale of a Master’s thesis. This thesis takes a modest but deliberate step into that space, treating the browser as a medium for embodied representation rather than as a delivery channel for visualisations that could as easily have been printed.

Taken together, the three traditions reviewed in this section support a claim that will structure the remainder of the thesis: the difficulty of communicating complex climate systems to non-specialist audiences is not addressable by refinement within the conventional-visualisation paradigm, and the direction in which the empirical, cognitive and design-research literatures collectively point is one that treats the reader as an active, embodied participant in the construction of understanding. The design of the three representational conditions in Chapter 7 is grounded in this claim, and the design decisions made there are defended through the perceptual and cognitive commitments this section has articulated.

3.5 Machine Learning for Climate and Earth-System Science

The four preceding sections established the representational problem this thesis addresses and the theoretical traditions in which its response is grounded. This section turns to the fifth strand of the literature: the applications of machine learning to climate and Earth-system science, which supply the analytical substrate on which the representational work of Chapter 7 is built. The characterisation offered here is not comprehensive; the field is now vast and has been surveyed elsewhere (Irrgang et al., 2021; Reichstein et al., 2019; Sonnewald et al., 2021). The purpose of this section is more specific: to identify the dominant agenda that has shaped the machine-learning-for-climate literature over the past decade and to locate, against that agenda, the alternative role for machine learning that this thesis undertakes.

The dominant agenda has been broadly *predictive* in orientation. The most cited applications of machine learning to Earth-system data fall into four families: probabilistic short-range forecasting of climate variables through recurrent and convolutional architectures (Sonnewald

et al., 2021; Weyn et al., 2020), anomaly detection and event characterisation in observational streams (Ham et al., 2019), subgrid-scale parameterisation of physical processes in numerical models (Brenowitz et al., 2020; Rasp et al., 2018) and statistical downscaling of coarse-resolution model output to regional scales (Vandal et al., 2019). The AMOC-specific literature reflects the same distribution: a growing body of work applies machine-learning techniques to detect early-warning signals in observational and reconstructed transport records (Ben-Yami et al., 2024; Boers, 2021; Ditlevsen & Ditlevsen, 2023), to forecast short-range variability from surface indicators (lu2024) and to identify latent dynamical modes in ensemble reanalyses (Sonnewald et al., 2019). The field has developed rapidly, has produced results of substantive scientific value and has established a set of methodological conventions — benchmark datasets, evaluation protocols and reporting standards — that are now recognisable across the community. The point relevant here is not that this agenda is misconceived; the point is that it is oriented, almost without exception, toward the production of *new scientific findings for expert audiences*, and that the very success of this orientation has left an adjacent question comparatively unexplored.

The adjacent question concerns what machine learning contributes when the goal is not to produce a new scientific result but to make an existing scientific object *intelligible to non-specialists*. The distinction is worth stating precisely because it is easy to lose. A recurrent neural network trained to forecast the AMOC transport strength six months in advance produces a new quantitative capability, evaluated against a ground-truth held-out record and reported in units — correlation coefficient, root-mean-square error, calibration score — that the climate-science community shares. A dimensionality-reduction analysis of the same record produces a latent representation whose value cannot be evaluated in those terms because it does not answer the same kind of question. It answers, instead, a question about *structure*: what are the small number of coordinates that account for most of the record’s variability, and what do those coordinates mean physically? The analytical work reported in Chapter 6 is of the second kind, and the design work of Chapter 7 rests on it. The claim that emerges is not that the AMOC vertical manifold has a novel low-dimensional structure — the finding is consistent with what oceanographers have understood about the meridional overturning for decades (Buckley & Marshall, 2016; Hannachi et al., 2007) — but that this structure, once recovered and characterised, can serve as the coordinate system through which the AMOC is made available to non-specialist perception. The role of machine learning here is not predictive but *representational*: it produces a substrate on which representational work can be conducted, rather than a forecast whose accuracy is the object of evaluation.

The literature that has treated machine learning in this second, representational role is comparatively thin, but it is not empty. Iten et al. (2020), working in fundamental physics rather than climate science, argued that the latent representations recovered by unsupervised networks can be treated as legitimate epistemological objects — as ways in which a computational system *proposes* a physical variable that a human interpreter can then examine and name. Camps-Valls et al. (2021) extended a related argument to seismology, using autoencoder representations of earthquake waveforms as the basis for interactive exploration by non-specialist audiences. In the visualisation-research community, **sacha2018** developed a framework for what they called *visual analytics with machine learning*, in which the model’s

latent representation is not the final product but the substrate through which the human analyst constructs understanding. The extension of this framework from expert analytics to public communication — from analyst interfaces to representational artefacts for non-specialists — has been proposed occasionally in the visualisation literature (Cabrera et al., 2023; Sacha et al., 2017) but has been implemented in the climate-communication literature almost nowhere. The specific proposition that this thesis pursues — that the latent representations recovered from an invisible climate system can be translated into experiential representations legible to non-specialist audiences — is coherent with the theoretical traditions reviewed in §3.4 and with the emerging visual-analytics tradition surveyed here, but its systematic empirical development in the climate-communication context has not yet been attempted. The gap this thesis addresses is defined at that intersection.

3.6 Synthesis: The Intersection This Thesis Occupies

Read as a single argument, the five strands of literature reviewed in this chapter converge on a proposition that no single strand articulates on its own. The psychological literature reviewed in §3.1 establishes that the reception of climate information is filtered through documented cognitive and cultural barriers which conventional formats not only fail to address but occasionally reinforce. The cognitive-science literature reviewed in §3.2 demonstrates that these barriers are compounded by structural limits on the human capacity to reason about accumulation, feedback and delay from static visual representations. The empirical visualisation research reviewed in §3.3 documents that the standard formats of climate communication have been evaluated primarily through preference studies rather than through comprehension studies, and that comprehension gains from alternative formats remain largely uncharted. The literatures on embodied cognition, data physicalisation and immersive analytics reviewed in §3.4 identify the theoretical and design traditions in which the response to this cluster of limitations has begun to take shape, but their application to climate communication has been sporadic and their systematic evaluation on specific climate systems almost absent. The machine-learning-for-climate literature reviewed in §3.5 has developed rapidly and with considerable technical sophistication, but its dominant orientation has been predictive rather than representational, and the possibility that machine-learning-derived latent structures might serve as the coordinate system for perceptually meaningful representation — rather than as inputs to a forecast — has not been pursued in the climate context.

The intersection at which these five convergences meet is narrow and empirically unpopulated. It is defined by the combination of five commitments: to a climate system whose invisibility and non-linearity make it a stress-test case for the representational problem (Armstrong McKay et al., 2022; Lenton et al., 2008), to a machine-learning treatment whose purpose is to recover latent structure rather than to forecast outcomes (Iten et al., 2020; Sonnewald et al., 2019), to a design tradition that treats the reader as an embodied and active participant in the construction of meaning (Jansen et al., 2015; Lakoff, n.d.; Marriott et al., 2018), to a comparative representational framework in which conventional and experiential formats can be evaluated against one another rather than each on its own terms (Harold

et al., 2016; Kause et al., 2021) and to a qualitative and exploratory evaluation methodology appropriate to a phenomenon whose measurement is not yet ripe for population-level inference (Braun & Clarke, 2006; Smith & McGannon, 2018). No study reviewed in the preceding sections satisfies all five commitments jointly, and the small number of studies that satisfy any three or four of them are distributed across the five disciplinary literatures rather than integrated within a single research programme. The gap this thesis addresses is not the absence of any one of these commitments in the literature; it is the absence of their integration.

The response this thesis makes to the situation identified in the preceding paragraph is deliberately modest in its scope and deliberately specific in its claims. It applies machine learning to the observational and reanalysis records of a single climate system, the Atlantic Meridional Overturning Circulation, in order to recover a latent structure whose faithfulness to the underlying dynamics is defensible on the terms the oceanographic literature itself recognises (Buckley & Marshall, 2016; Hannachi et al., 2007). It designs three representational conditions — conventional static, interactive dynamic and experiential — through which that latent structure is translated for non-specialist audiences, and it grounds each design decision in the perceptual and cognitive principles identified in §3.4. It evaluates the three conditions through a small-N think-aloud study whose epistemic ambition is the qualitative depth appropriate to a first empirical investigation, not the population-level generalisation that would require a study design outside the scope of this work. The contribution offered is not a novel finding in any of the five literatures reviewed in this chapter but a methodology that connects them, and the case for the value of that methodology rests on the argument that has now been made. The chapters that follow are its execution.

Chapter 4

Background: The Physical and Mathematical Foundation

[Port the existing Chapter 4 from the Word working document. This is the chapter you have already drafted in depth: Navier–Stokes, the AMOC as a meridional volume transport integral, the Stommel model and bistability, the theoretical basis of Early Warning Signals.]

Part II

Translation: From Data to Experience

Chapter 5

Methodology

[To be drafted in Week 3. Three-part methodology: (1) Analytical methods (PCA, autoencoder, IF, EWS) — already executed; (2) Translation methods (design grammar, three representational conditions C1/C2/C3) — to be specified; (3) Evaluation methods (think-aloud protocol, thematic analysis per Braun & Clarke 2006) — to be specified.]

Chapter 6

Detect: Analytical Pipeline and Results

[Port the current Chapters 5–6 from the Word document. This is the analytical results chapter: data preprocessing, PCA, autoencoder, EWS, anomaly detection. The full validation discussed in the live session lives here, with the three null results framed as foundations of Part II.]

Chapter 7

Translate: Designing Experiential Representations

[To be drafted in Weeks 4–6. The design grammar (DP2.1, DP2.2), the three representational conditions, the technical implementation of condition C3 (Three.js + Tone.js + D3), and the documentation of every perceptual mapping decision with its theoretical justification.]

Part III

Evaluation & Reflection

Chapter 8

Evaluate: Designing and Conducting the User Study

[To be drafted in Week 7. The qualitative exploratory study: participant recruitment, think-aloud protocol, ethical considerations, analytical framework (thematic analysis).]

Chapter 9

Results

[To be drafted in Week 8–9. Analytical results from Part I (already obtained), design results from Part II (the realised prototypes), empirical results from Part III (the user study findings).]

Chapter 10

Discussion

[To be drafted in Week 9. Interpretation of the analytical null results (rigour as contribution), the design propositions as realised artefacts, the user-study findings in conversation with the literature on cognition and climate communication, and reflection on the limitations of the methodology.]

Chapter 11

Conclusion and Future Work

[To be drafted in Week 9. Restatement of the contribution, honest scope claims, explicit future work (LSTM forecasting, extension to other tipping elements such as the Greenland Ice Sheet as a fully analytical case study).]

11.1 Reproducibility and Open Science

The scientific claims made in this thesis are grounded in analyses that can be reproduced in full from open sources. The RAPID observational data used throughout Part II is publicly distributed by the British Oceanographic Data Centre (Ben I Moat et al., 2026); the Copernicus Marine Service reanalysis ensemble is accessible under the Copernicus data licence. The analytical pipeline that produced every figure and quantitative result reported in Chapter 6 is maintained as a public repository, with a fixed conda environment specification, versioned notebooks and a documented workflow: <https://github.com/enricovaccari/amoc-thesis-pipeline>. The writing of this thesis, including its $\text{\LaTeX}$ sources and its bibliographic database, is maintained as a separate public repository at <https://github.com/enricovaccari/amoc-thesis-writing>. The commitment underlying this openness is that a scientific claim which cannot be independently verified is, in a meaningful sense, not a scientific claim; the technical documentation required to reproduce the results reported here is provided in full in Appendix 11.1.

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Appendix A — Ethics Documentation

[Consent form, participant information sheet, data-handling protocol. To be drafted in Week 3.]

Appendix B — Study Protocols

[Think-aloud script, evaluation prompts, post-task interview schedule. To be drafted in Week 3.]

Appendix C — Code and Reproducibility

[Pointer to the GitHub repository, key code listings, environment specification, instructions to reproduce the analytical pipeline.]